Inefficient Nonsense Mediated Decay In Murine Cre/Lox-Mediated Conditional Knockouts

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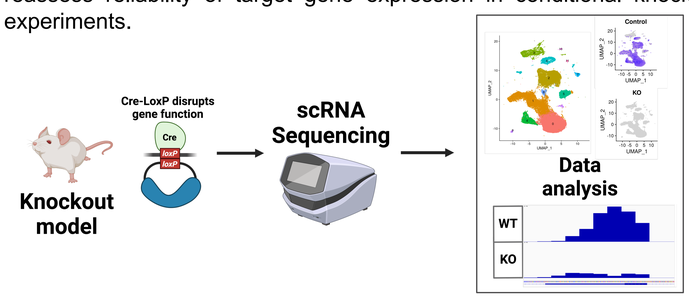
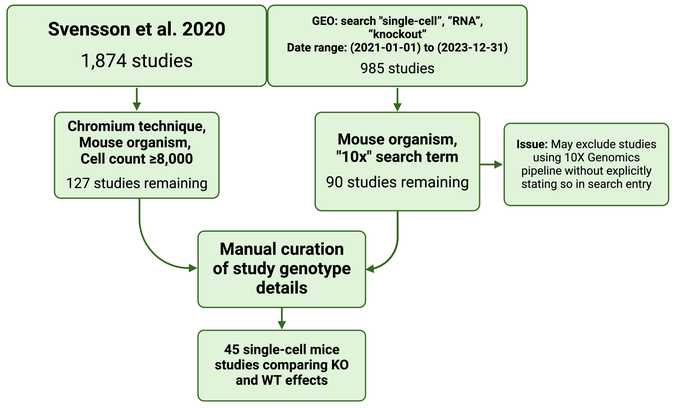
Introduction

- Mice knockout models often involve removing exons in target organism transcripts, creating frameshift mutations that trigger **nonsense-mediated decay (NMD)**[1].
- Recent evidence shows NMD efficiency can vary -- some truncated transcripts may escape degradation[2]. This concept challenges the assumption of null alleles in knockout models.
- **Hypothesis: Murine exon deletion models can retain target mRNA expression due to varying NMD efficiency.**
- **Aim:** Analyze single-cell RNA-seq data across public sources to reassess reliability of target gene expression in conditional knockout experiments.



Methods

We filtered 2,859 single-cell transcriptomic studies to analyze gene knockout effects in mice, comparing knockout (KO) and wild-type (WT) genotypes across cell clusters – resulting in 45 studies for our database.



Key analytical steps:

1. Apply study-specific thresholds for RNA features, counts, and mitochondrial gene percentage for clustering. Identify clusters with significant gene expression using 5th percentile threshold of overall expression.
2. Compare average expression between genotypes in high-expressing clusters and calculate NMD efficiency as per the formula below[3].

$$\text{NMD efficiency} = -\log_2 \frac{KO\ AverageExpression}{WT\ AverageExpression}$$

Results

Variability in NMD and Knockout Efficiency Across All Studies

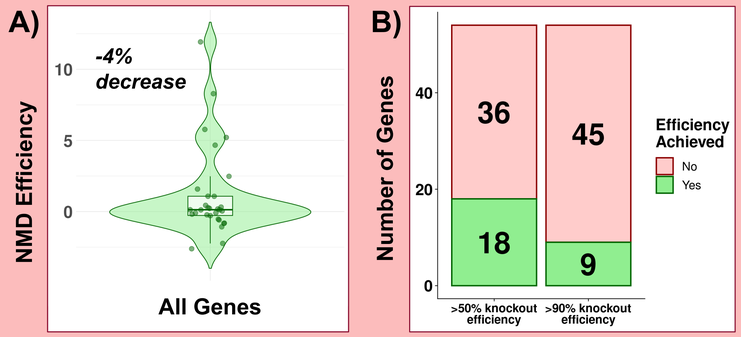


Figure 1. A) Distribution of relative knockdown percentages resulting from premature termination codons (PTCs) and **B)** comparison of knockout efficiency with prespecified thresholds illustrate the variability in NMD efficiency across all observed genes.

PTC Position Fails to Predict mRNA Persistence in Conditional Knockouts

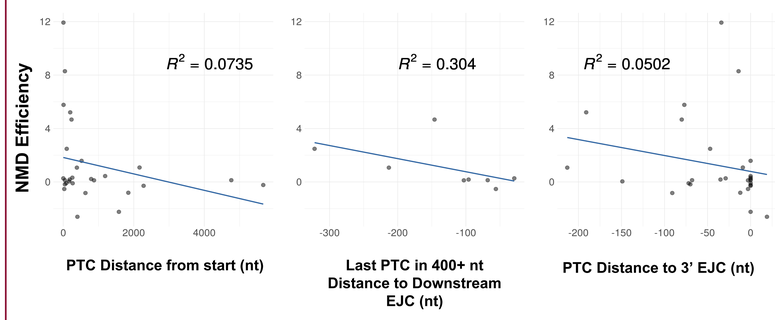


Figure 2. Scatterplots showing the relationship between PTC location (distance from start, distance from long exon, and distance to 3' exon junction complex) and relative NMD efficiency. Low correlation demonstrates that commonly-recognized NMD efficiency rules[3] do not consistently predict transcript levels in single-cell data from conditional knockouts on a continuous scale.

Classic NMD Rules Poorly Predict mRNA Levels in Conditional Knockouts

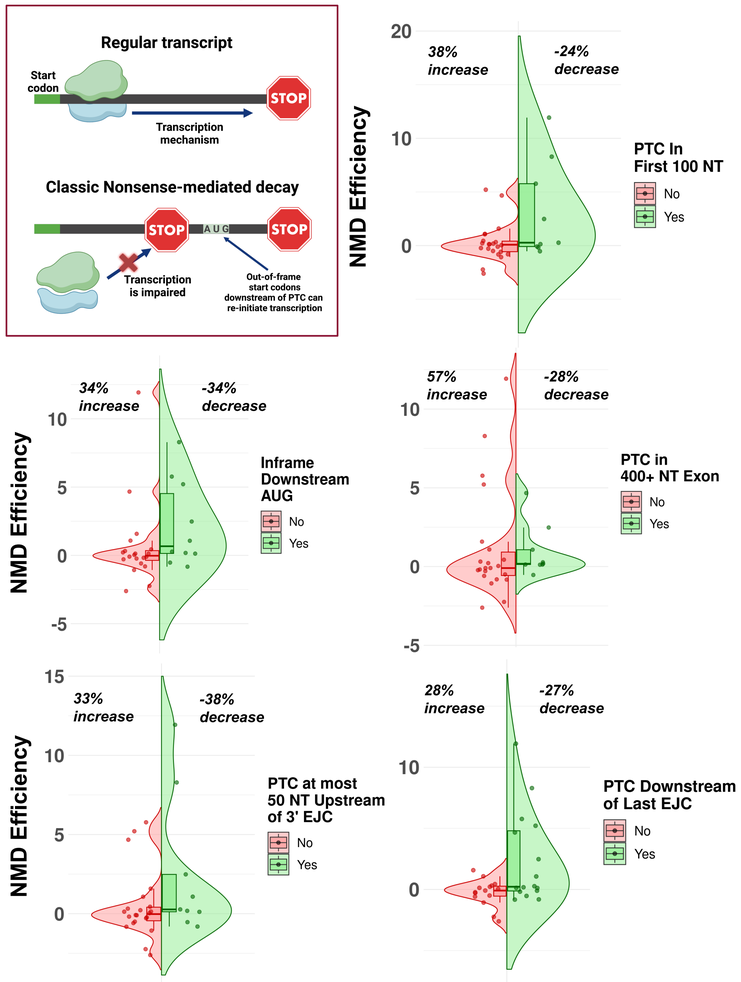


Figure 3. Violin plots comparing NMD efficiency among conditional knockout models using classic NMD rules[3] reveal substantial overlap in NMD efficiency distributions for each condition, suggesting that these traditional predictors of NMD efficiency fail to accurately forecast mRNA levels in conditional knockout models.

Conclusions

- NMD in murine conditional knockouts appears to be **less efficient than commonly recognized**.
- Classic "rules" for NMD efficiency, such as PTC location relative to exon junctions, **do not consistently predict** the actual changes in transcript levels observed in single-cell data.

References

1. Brogna, S., & Wen, J. (2009). Nonsense-mediated mRNA decay (NMD) mechanisms. Nature Structural & Molecular Biology (Vol. 16, Issue 2, pp. 107–113).
2. Cheng, P. et al. (2022). Smad3 regulates smooth muscle cell fate and mediates adverse remodeling and calcification of the atherosclerotic plaque. Nature Cardiovascular Research (Vol. 1, Issue 4, pp. 322–333).
3. Lindeboom, R. G. H., et al. (2016). The rules and impact of nonsense-mediated mRNA decay in human cancers. Nature Genetics (Vol. 48, Issue 10, pp. 1112–1118).

Example schematics created using BioRender.com.